



Dayananda Sagar University School of Engineering

Devarakaggalahalli, Harohalli Kanakapura Road, Bangalore South Dt., Karnataka 562112

DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY

B.Tech Final Year Capstone Project Phase 1-Weekly Progress Report

Project Semester: 7

Aug-Dec 2026

Week No: 5

Date : From : 31/08/2026 To 04/09/2026

Department	Computer Science and Technology
Team Number	Team 3
Names & USN of Team Members	Putta Sujith & ENG23CT0058 K Vikas Aneesh Reddy & ENG23CT0052 Karnati Mokshith & ENG23CT0053
Name of the Guide	Dr. Ramandeep Kaur
Project Title	IntentCloud - Intent-Aware Cognitive Cloud Memory System

Work Carried out by the team during the Week (in Bullet points only):

- Implemented sparse/BM25-style keyword scoring alongside dense retrieval.
- Implemented Reciprocal Rank Fusion (RRF) to merge dense and sparse rankings.
- Integrated Qdrant hybrid search functionality for combined retrieval.
- Added the cross-encoder reranking step to reorder the retrieved candidates and return the final top-3 results.
- Updated the Next.js /search page to display reranked top-3 results with highlighted matched snippets.
- Prepared the 30–50 query evaluation set with independently authored ground-truth target files.
- Placed the Raspberry Pi hardware order for the memory-layer implementation.

Individual Contribution from the Student:

Putta Sujith	● Implemented sparse/BM25-style keyword scoring alongside dense retrieval. ● Implemented Reciprocal Rank Fusion (RRF) to merge dense and sparse rankings before reranking. ● Tested the combined retrieval flow with different natural-language queries. ● Verified that the hybrid retrieval pipeline improves candidate selection compared with dense-only retrieval.
K Vikas Aneesh Reddy	● Integrated Qdrant's native hybrid query functionality for dense and sparse retrieval.

	● Added the cross-encoder (ms-marco-MiniLM) reranking step over the fused top-N candidates. ● Tested the reranking pipeline and verified the final top-3 result generation. ● Placed the Raspberry Pi hardware order for the upcoming memory-layer deployment.
Karnati Mokshith	● Updated the Next.js /search page to display the reranked top-3 search results. ● Added highlighted matched-snippet explanations for the retrieved documents. ● Tested the updated search interface with hybrid and reranked results. ● Prepared the 30–50 query evaluation set with independently authored ground-truth target files.
Work Planned for the Following Week: ● Prepare and deliver the Review-2 demonstration with the hybrid and reranked retrieval pipeline. ● Flash Raspberry Pi OS (64-bit, Bookworm) and complete the headless SSH setup. ● Install Python, Qdrant, and FastAPI on the Raspberry Pi. ● Migrate the Qdrant index and file storage from the laptop to the Raspberry Pi. ● Verify communication between the Raspberry Pi and laptop over the local network. ● Tune the confidence threshold for the “no confident match” response. ● Finalize and submit the research paper by 12 September 2026.	

Percentage Work Completed: 48%

Signature of the Students:

Signature of the Guide(s)